

Department of Computer and Software Engineering**SE: Machine Learning****Course Instructor:** Dr. Ahmed Raza**Date:****Day:****Semester:****Lab 5: Decision Tree Split using Gini and Information Gain**

Lab Title	CLO	BT	PLO	Weightage
Decision Tree Split using Gini and Information Gain				

Name	Roll Number	Report /10	Viva /5	Total /15
Zunaira Abdul Aziz	BSSE23058			

Checked on: _____

Signature: _____

1 Learning Outcomes

After completing this lab, the student will be able to:

- Compute Gini impurity for binary classification.
- Calculate Information Gain from candidate splits.
- Implement the split selection procedure used by decision trees.
- Observe bias toward high-cardinality features.

2 Dataset

We will use the Iris dataset from sklearn. To simplify the problem we restrict the dataset to two classes.

Properties:

- Samples: 100
- Features: 4
- Classes: Binary

Dataset loading:

```
from sklearn.datasets import load_iris
```

3 Gini Impurity

For binary classification:

$$Gini = 1 - (p_0^2 + p_1^2)$$

A perfectly pure node has Gini impurity equal to 0.

4 Information Gain

The reduction in impurity after a split is:

$$Gain = Gini(parent) - \frac{n_L}{n} Gini(L) - \frac{n_R}{n} Gini(R)$$

The split with the highest information gain is selected.

5 Part A: Implement Gini Impurity (3 Marks)

Implement the function:

`gini(y)`

The function should:

- compute class probabilities
- apply the Gini formula
- return impurity

Verify using:

`y = [0,0,0,1]`

Compute impurity manually and compare.

```
def gini(y):  
    if len(y) == 0:  
        return 0  
    p0 = np.mean(y == 0)  
    p1 = np.mean(y == 1)  
  
    return 1 - (p0**2 + p1**2)
```

6 Part B: Implement Information Gain (4 Marks)

Implement:

`information_gain(parent, left, right)`

Steps:

- compute parent impurity
- compute child impurities
- compute weighted average
- subtract from parent impurity

```
def information_gain(parent, left, right):
    parent_impurity = gini(parent)
    n = len(parent)
    n_left, n_right = len(left), len(right)

    if n == 0:
        return 0
    weighted_child_impurity = (n_left / n) * gini(left) + (n_right / n)
    * gini(right)

    return parent_impurity - weighted_child_impurity
```

7 Part C: Implement Tree Split Logic (4 Marks)

Implement the function:

`find_best_split(X, y)` The

function should:

- iterate over features
- compute candidate thresholds
- evaluate information gain
- return the best feature and threshold Use

NumPy operations where possible.

```
def find_best_split(X, y):
    best_gain = -1
    best_feature = None
    best_threshold = None

    n_features = X.shape[1]

    for feature_idx in range(n_features):
        values = X[:, feature_idx]
        thresholds = candidate_thresholds(values)

        for threshold in thresholds:
            _, _, y_left, y_right = split_dataset(X, y, feature_idx,
threshold)
            gain = information_gain(y, y_left, y_right)
            if gain > best_gain:
                best_gain = gain
                best_feature = feature_idx
                best_threshold = threshold

    return best_feature, best_threshold, best_gain
```

8 Part D: Cardinality Bias Experiment (4 Marks)

A new feature will be added:

```
index_feature = sample_index / N
```

This feature contains many unique values. Run the split algorithm again and observe:

- which feature is selected

In both the original and augmented datasets, Feature 2 (Petal Length) was selected as the best split with a threshold of 2.45 and an Information Gain of 0.49875

- whether the high-cardinality feature is preferred

No. In this specific run, the high-cardinality index feature was not preferred over the physical features of the Iris dataset.

Explain why this occurs.

- **Cardinality Bias:** Information Gain mathematically tends to favor features with many unique values (high cardinality) because they can create very pure child nodes by isolating small groups of samples.
- **Overfitting Risk:** If the algorithm had selected the index feature, it would be memorizing the specific order of the training data rather than finding a general biological pattern, leading to high variance and poor performance on new data.
- **Result Interpretation:** The reason it wasn't selected in this code is likely that Feature 2 provides a near-perfect split for the binary classes (Iris-Setosa vs. Iris-Versicolor) with a single threshold, yielding a higher gain than a single split on a high-cardinality index.

9 Expected Output

Your program should print:

- Best split feature

- Threshold
- Information gain
- Result of cardinality bias experiment

```
Original Dataset  
Best Feature: 2  
Threshold: 2.45  
Information Gain: 0.49875
```

```
With High Cardinality Feature  
Best Feature: 2  
Threshold: 2.45  
Information Gain: 0.49875
```

Assessment Rubrics

Method: Lab reports and instructor observation during lab sessions

Outcome Assessed:

1. Ability to conduct experiments, as well as to analyze and interpret data (P).
2. Ability to use the techniques, skills, and modern engineering tools necessary for engineering practice (P).

Performance	Exceeds expectation (4-5)	Meets expectation (3-2)	Does not meet expectation (1)	Marks
1. Realization of Experiment [a, b]	Selects relevant equipment to the experiment, develops setup diagrams of equipment connections or wiring.	Needs guidance to select relevant equipment and to develop equipment connection or wiring diagrams.	Incapable of selecting relevant equipment to conduct the experiment.	
2. Conducting Experiment [a, b]	Does proper calibration of equipment, carefully examines equipment moving parts, and ensures smooth operation.	Calibrates equipment, examines equipment moving parts, and operates the equipment with minor error.	Unable to calibrate appropriate equipment, and equipment operation is substantially wrong.	
3. Laboratory Safety Rules [a]	Respectfully and carefully observes safety rules and procedures.	Observes safety rules and procedures with minor deviation.	Disregards safety rules and procedures.	
6. Data Collection [a]	Plans data collection to achieve experimental objectives, and conducts an orderly and complete data collection.	Plans data collection to achieve experimental objectives, and collects complete data with minor error.	Does not know how to plan data collection; data collected is incomplete and contain errors.	
7. Data Analysis [a]	Accurately conducts simple computations and statistical analysis; correlates results to known theoretical values; accounts for measurement errors.	Conducts simple computations with minor error; reasonably correlates results to known theoretical values; attempts to account for errors.	Unable to conduct simple statistical analysis; no attempt to correlate or explain errors.	
8. Computer Use [a]	Uses computer to collect and analyze data effectively.	Uses computer to collect and analyze data with minor error.	Does not know how to use computer to collect and analyze data.	
Total				

Faculty:

Name: _____

Signature: _____

Date: _____